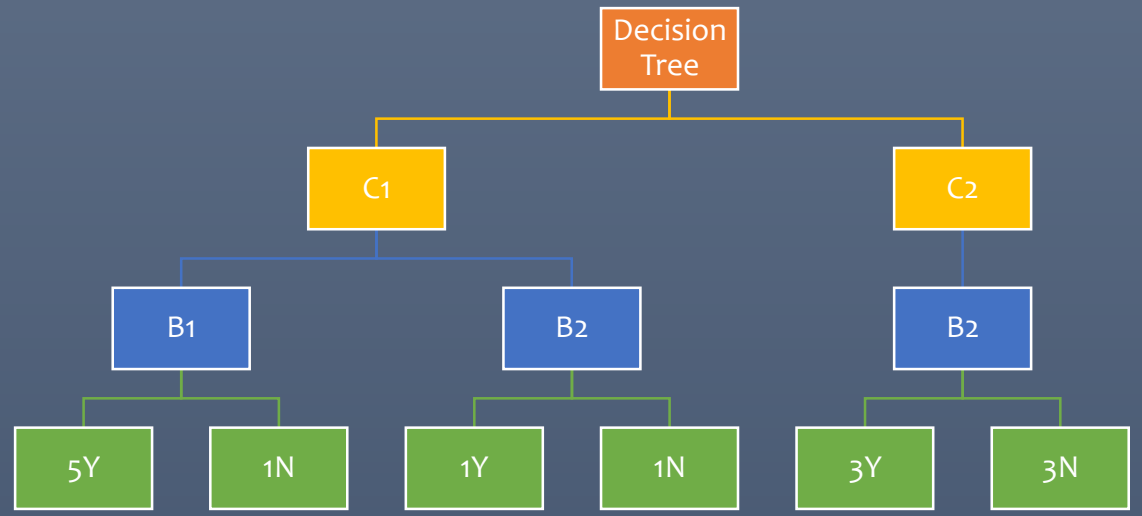
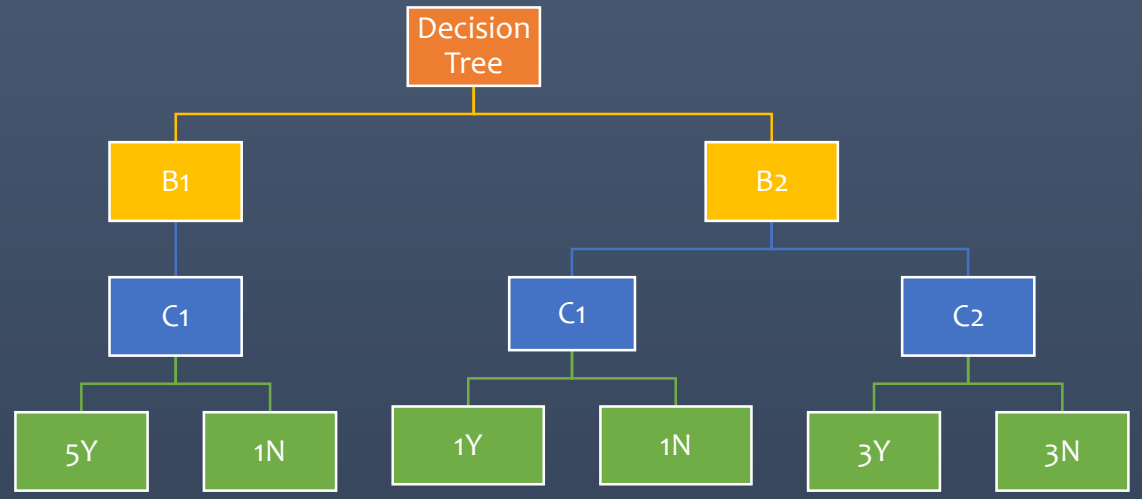
Feature 01	Feature 02	Yes/No
C1	*B1*	Y
C1	*B1*	Y
C1	*B1*	Y
C1	*B1*	Y
C1	*B1*	Y
C1	B2	Y
C2	B2	Y
C2	B2	Y
C2	B2	Y
C1	*B1*	N
C1	B2	N
C2	B2	N
C2	B2	N
C2	B2	N



When constructing the tree, the model must choose one feature among multiple features as the parent node. This selection is based on the concept of **information gain**, where the feature with the highest information gain becomes the parent node, while the remaining features become the child nodes.



It is basically **splitting the data into a tree-like structure**.
A Decision Tree has three main parts:

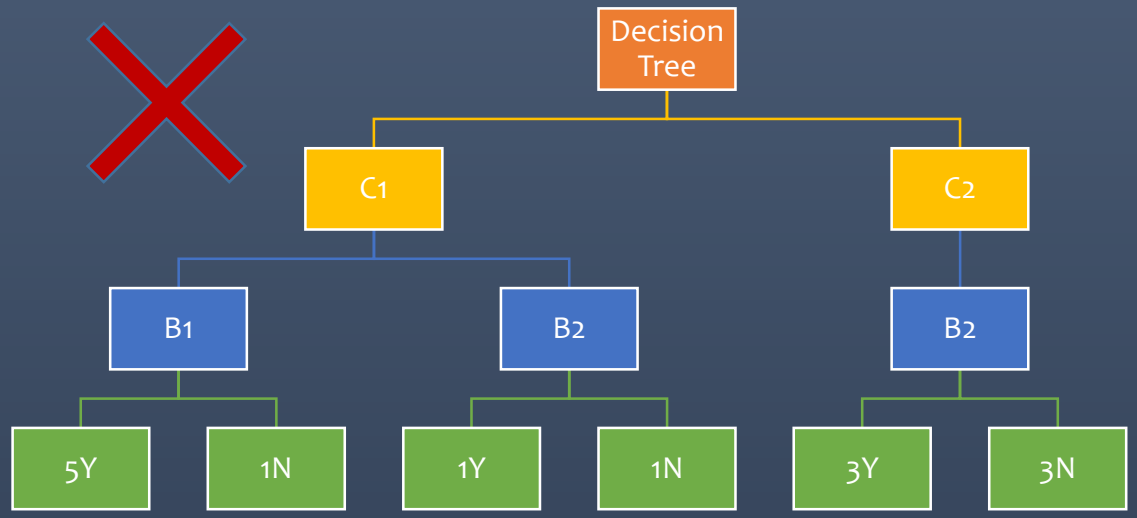
- **Root Node** – the starting point of the tree.
- **Child Nodes** – the branches created after splitting the data.
- **Leaf Nodes** – the final result or prediction.

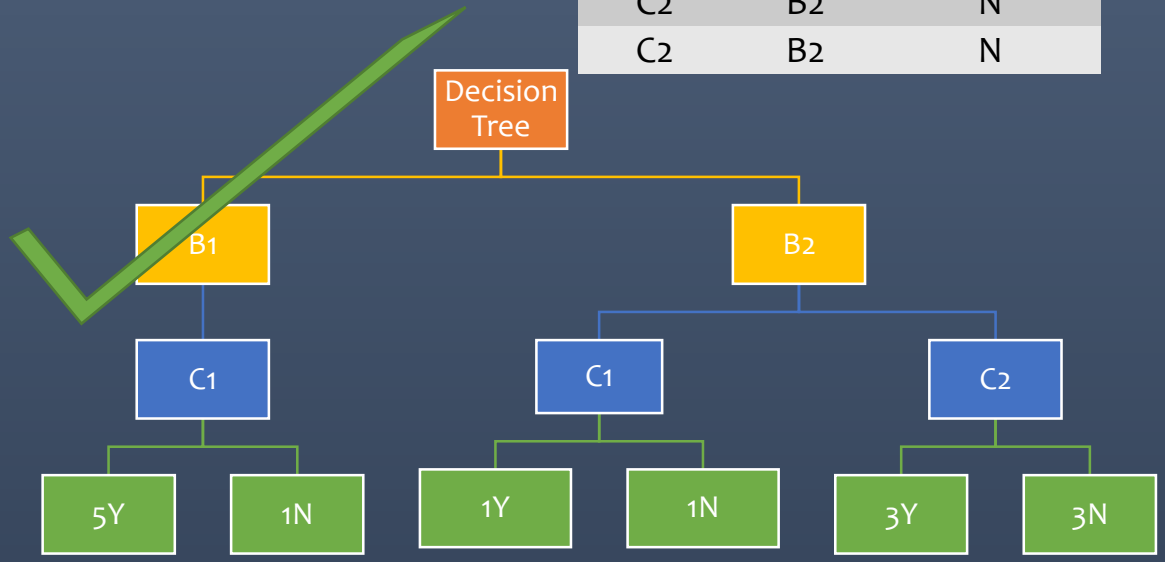
Since there are multiple features in the dataset, we need to decide **which feature should be the Root Node** and which features should come next as Child Nodes.
The feature with the **highest Information Gain** is selected as the **Root Node.** The other features are then used for the subsequent child nodes.
Steps to Calculate Information Gain

1. First, calculate **Entropy** or **Gini Impurity**.
2. Use the calculated value in the **Information Gain formula**.
3. Calculate the Information Gain for **each feature**.
4. Compare the Information Gain of all features.
5. The feature with the **highest Information Gain becomes the Root Node**.
6. The remaining features are used to create the **subsequent Child Nodes**.

	Information Gain	
	Entropy	Gini Impurity
Feature One	0.049	0.029
Feature Two	0.090	0.074

Feature 01	Feature 02	Yes/No
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B2	Y
C2	B2	Y
C2	B2	Y
C2	B2	Y
C1	B1	N
C1	B2	N
C2	B2	N
C2	B2	N
C2	B2	N



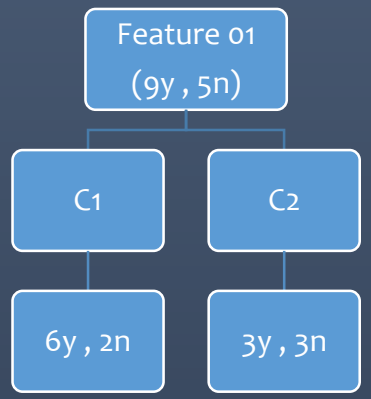


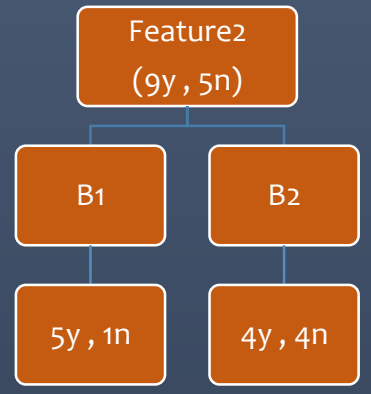
How to calculate the Information gain using

- Entropy
- Gini Impurity

Feature 01	Yes/No
C1	Y
C1	Y
C1	Y
C1	Y
C1	Y
C1	Y
C2	Y
C2	Y
C2	Y
C1	N
C1	N
C2	N
C2	N
C2	N

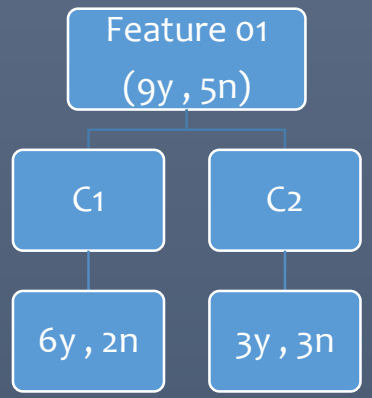
Feature 02	Yes/No
B1	Y
B1	Y
B1	Y
B1	Y
B1	Y
B2	Y
B2	Y
B2	Y
B2	Y
B1	N
B2	N
B2	N
B2	N
B2	N





Calculate the Entropy of Feature One

$$Entroy = -\sum_{I=1}^{N} p_i \, log_2 \, p_i$$



Entropy of Root Node(9y , 5n)

$$Entroy = (-\mathrm{P}_y) * (\log_2(\mathrm{P}_y)) + (-\mathrm{P}_n) * (\log_2(\mathrm{P}_n))$$

$$= \left(-\left(\frac{9}{14}\right)\right) * \left(\log_2\left(\frac{9}{14}\right)\right) + \left(-\left(\frac{5}{14}\right)\right) * \left(\log_2\left(\frac{5}{14}\right)\right)$$

$$= (-0.64) * (\log_2(0.64) + (-0.35) * (\log_2(0.35))$$

$$= (-0.64) * (-0.64) + (-0.35) * (-1.49)$$

$$\boldsymbol{Entroy = 0.94}$$

Entropy of Left Node (6y , 2n)

$$Entroy = -\mathrm{P}_y \log_2(\mathrm{P}_y) - \mathrm{P}_n \log_2(\mathrm{P}_n)$$

$$= \left(-\left(\frac{6}{8}\right)\right) * \left(\log_2\left(\frac{6}{8}\right)\right) + \left(-\left(\frac{2}{8}\right)\right) * \left(\log_2\left(\frac{2}{8}\right)\right)$$

… … … … … … .

$$\boldsymbol{Entroy = 0.81}$$

Entropy of a Right Node(3y , 3n)

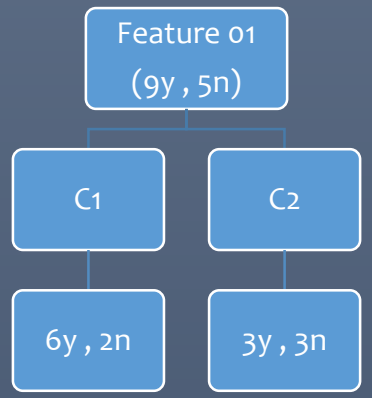
$$Entroy = -\mathrm{P}_y \log_2(\mathrm{P}_y) - \mathrm{P}_n \log_2(\mathrm{P}_n)$$

$$= \left(-\left(\frac{3}{6}\right)\right) * \left(\log_2\left(\frac{3}{6}\right)\right) + \left(-\left(\frac{3}{6}\right)\right) * \left(\log_2\left(\frac{3}{6}\right)\right)$$

… … … … … … .

$$\boldsymbol{Entroy = 1}$$

Gini Impurity the Entropy of Feature One



$$Gini = 1 - \sum_{i=1}^{n} (p_i\ )^2$$

GI of Root Node(9y , 5n)

$$Gini = 1 - \left[\left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2 \right]$$

$$= 1 - [\ (0.64)^2 + (0.36)^2\ ]$$

$$= 1 - [\ 0.41 + 0.13\ ]$$

$$\boldsymbol{Gini} = \mathbf{0.46}$$

GI of Left Node (6y , 2n)

$$Gini = 1 - \left[\left(\frac{6}{8}\right)^2 + \left(\frac{2}{8}\right)^2 \right]$$

$$= 1 - [\ (0.75)^2 + (0.25)^2\ ]$$

$$= 1 - [\ 0.56 + 0.06\ ]$$

$$\boldsymbol{Gini} = 0.38$$

GI of Right Node(3y , 3n)

$$Gini = 1 - \left[\left(\frac{3}{6}\right)^2 + \left(\frac{3}{6}\right)^2 \right]$$

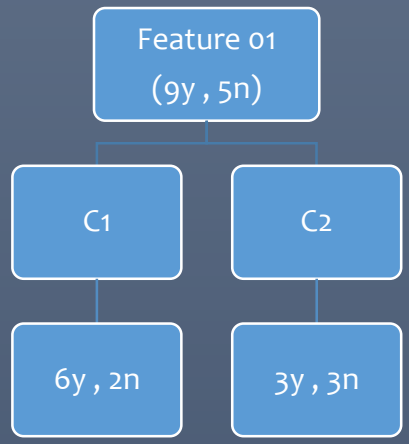
$$= 1 - [\ (0.50)^2 + (0.50)^2\ ]$$

$$= 1 - [\ 0.25 + 0.24\ ]$$

$$\boldsymbol{Gini} = 0.50$$

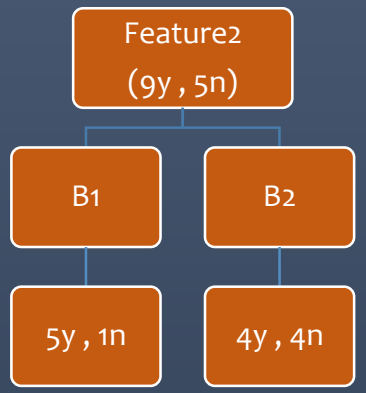
Entropy and Gini Impurity the Entropy of Feature One

	Root Node	Child Node	Child Node
Feature One	(9y , 5n)	(6y , 2n)	(3y , 3n)
Entropy	0.94	0.81	1
Gini Impurity	0.46	0.38	0.5



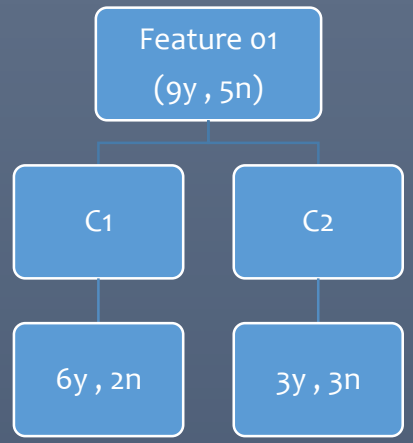
Entropy and Gini Impurity of Feature Two

	Root Node	Child Node	Child Node
Feature Two	(9y , 5n)	(5y , 1n)	(4y , 4n)
Entropy	0.94	0.65	1
Gini Impurity	0.48	0.28	0.5



Calculate the Information Gain of the Feature One

	Root Node	Child Node	Child Node
Feature One	(9y , 5n)	(6y , 2n)	(3y , 3n)
Entropy	0.94	0.81	1
Gini Impurity	0.46	0.38	0.5



$$Information\ Gain(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

H(S) = Entropy or Gini of the root Node.
Sv = total values in a particular node.
S = Total Number of Values of Output Feature
H(Sv) = Entropy or Gini of the Child Node

IG Using Entropy

$$Gain(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

$$Gain(S, f1) = H(S) - \left[\left(\frac{|Sv|}{|S|} * H(Sv)\right) + \left(\frac{|Sv|}{|S|} * H(Sv)\right)\right]$$

$$Gain(S, f1) = 0.94 - \left[\left(\frac{8}{14} * 0.81\right) + \left(\frac{6}{14} * 1\right)\right]$$

$$Gain(S, f1) = 0.049$$

IG Using GI

$$Gain(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

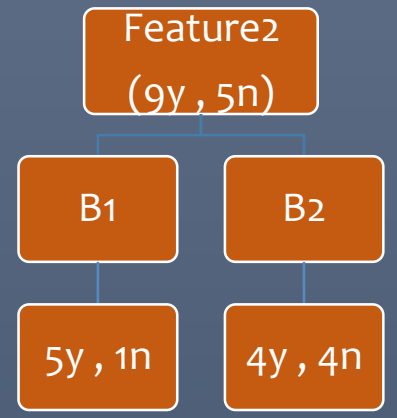
$$Gain(S, f1) = H(S) - \left[\left(\frac{|Sv|}{|S|} * H(Sv)\right) + \left(\frac{|Sv|}{|S|} * H(Sv)\right)\right]$$

$$Gain(S, f1) = 0.46 - \left[\left(\frac{8}{14} * 0.38\right) + \left(\frac{6}{14} * 0.50\right)\right]$$

$$Gain(S, f1) = 0.029$$

Calculate the Information Gain of the Feature Two

	Root Node	Child Node	Child Node
Feature Two	(9y , 5n)	(5y , 1n)	(4y , 4n)
Entropy	0.94	0.65	1
Gini Impurity	0.48	0.28	0.5



$$Information\ Gain(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

H(S) = Entropy or Gini of the root Node.
Sv = total values in a particular node.
S = Total Number of Values of Output Feature
H(Sv) = Entropy or Gini of the Child Node

IG Using Entropy

$$Gain(S, f2) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

$$Gain(S, f2) = H(S) - \left[\left(\frac{|Sv|}{|S|} * H(Sv)\right) + \left(\frac{|Sv|}{|S|} * H(Sv)\right)\right]$$

$$Gain(S, f2) = 0.94 - \left[\left(\frac{6}{14} * 0.65\right) + \left(\frac{8}{14} * 1\right)\right]$$

$$Gain(S, f2) = 0.090$$

IG Using GI

$$Gain(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

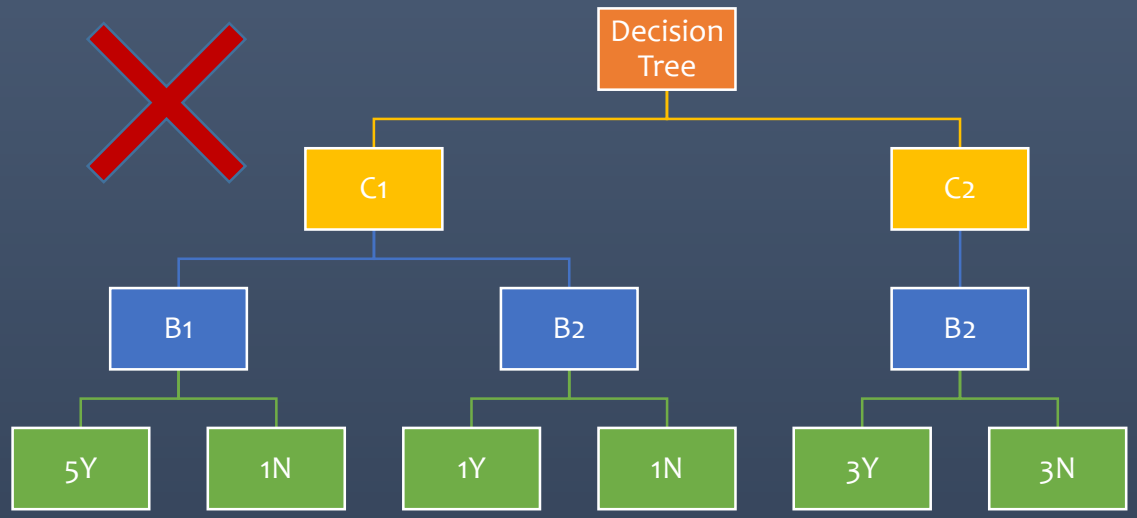
$$Gain(S, f1) = H(S) - \left[\left(\frac{|Sv|}{|S|} * H(Sv)\right) + \left(\frac{|Sv|}{|S|} * H(Sv)\right)\right]$$

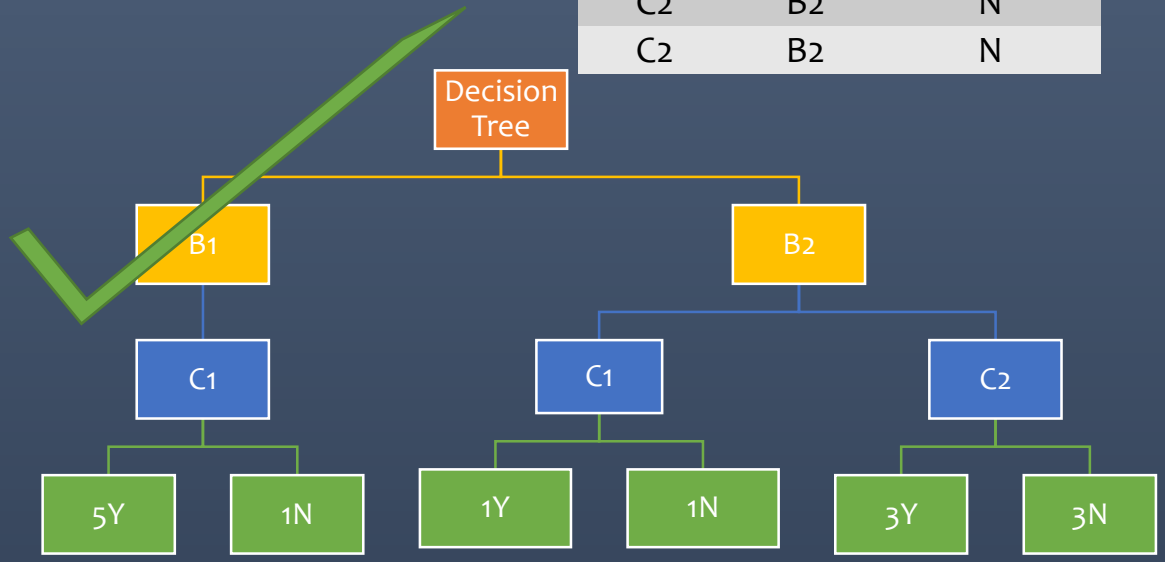
$$Gain(S, f1) = 0.46 - \left[\left(\frac{6}{14} * 0.28\right) + \left(\frac{8}{14} * 0.50\right)\right]$$

$$Gain(S, f1) = 0.074$$

	Information Gain	
	Entropy	Gini Impurity
Feature One	0.049	0.029
Feature Two	0.090	0.074

Feature 01	Feature 02	Yes/No
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B2	Y
C2	B2	Y
C2	B2	Y
C2	B2	Y
C1	B1	N
C1	B2	N
C2	B2	N
C2	B2	N
C2	B2	N





Hyper parameter

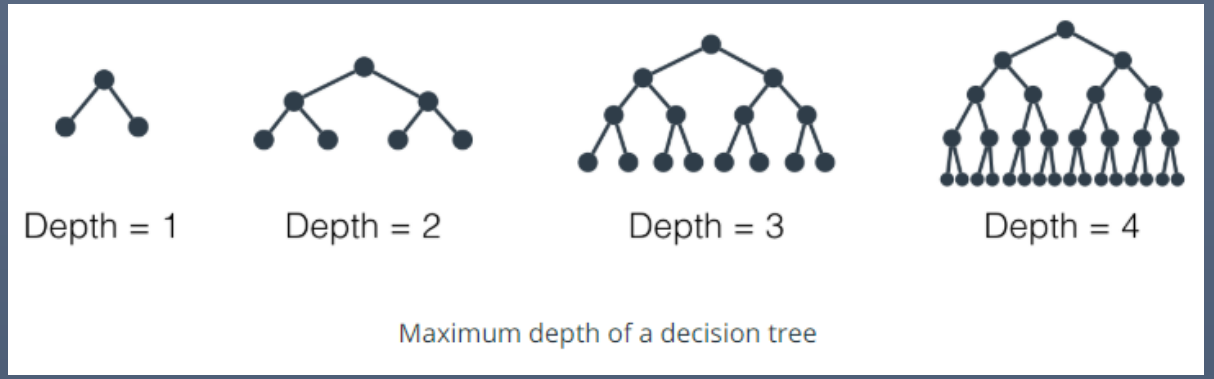
Criterion Gini impurity or Entropy

max Features: Number of feature to be short listed for model trainining

Example if you have 50 feature and for model training you keep only 10 feature as max feature you model will be trained on 10 features

Hyper parameter

Max Depth
Min Sample Split:
Min Sample Leaf:



Maximum depth of a decision tree

Hyper parameter tuning.

Max Depth:-

The height of a decision tree, also known as its depth, represents the number of levels or layers it contains.

Min Sample Split:

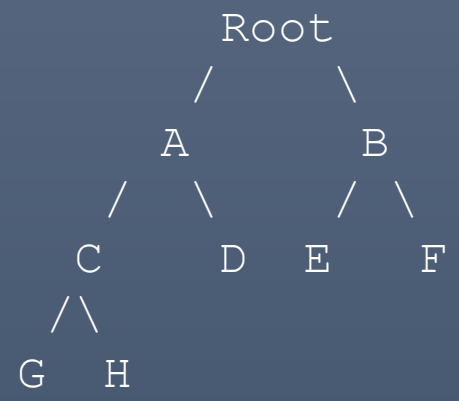
Minimum sample split decides minimum number of samples necessary to split a **nonterminal node.**
By default, the decision tree tries to split every node with two or more.

Min Sample Leaf:

it defines the minimum number of **data samples required to form a terminal leaf node** in a decision tree.

Pruning
in a Decision Tree means **removing unnecessary branches/splits from a tree to reduce overfitting.**

The tree may become extremely complex and start **memorizing the training data** rather than learning general patterns.



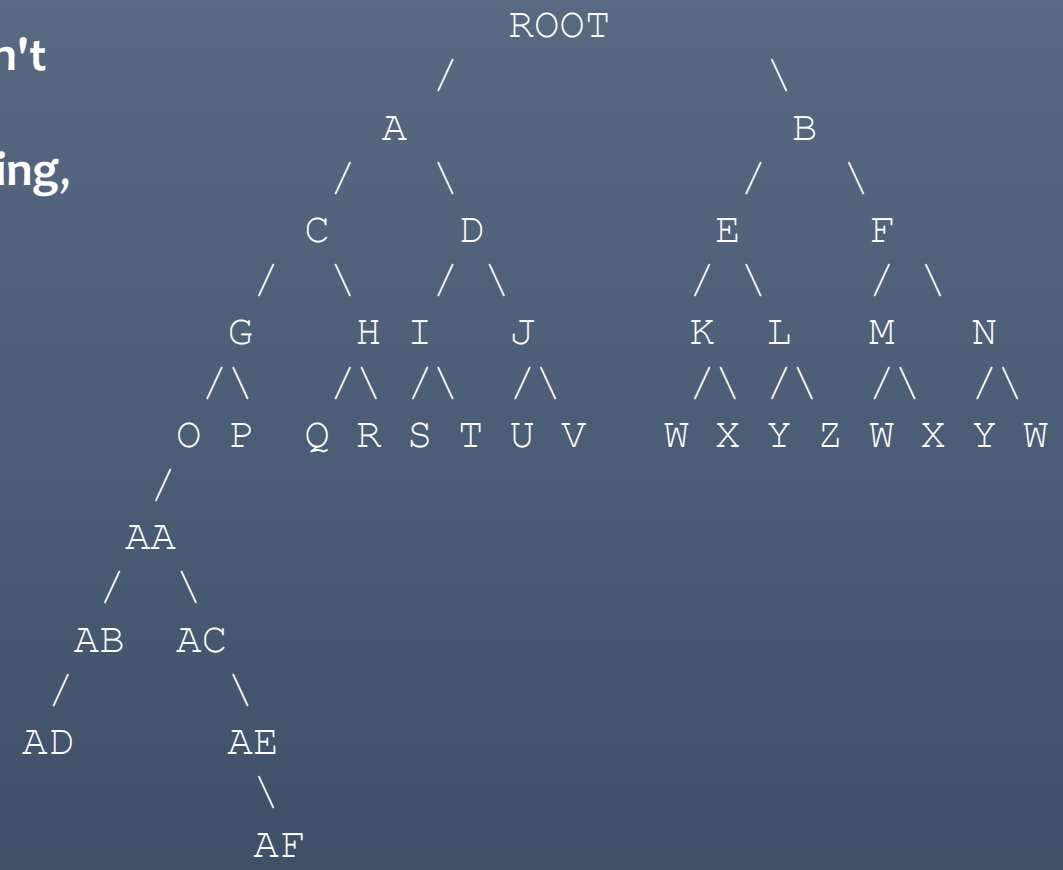
Max Depth
Min Sample Split:
Min Sample Leaf:

Post Pruning:

First, you allow the tree to grow relatively large:

Then you examine the tree and **remove branches that aren't providing enough benefit**.

In scikit-learn, a common method is **Cost-Complexity Pruning,** controlled by: ccp_alpha



DecisionTreeClassifier(ccp_alpha=0.0)

↓

Very large tree

↓

Potential overfitting

DecisionTreeClassifier(ccp_alpha=0.01)

↓

Some branches removed

↓

May be best fit model

DecisionTreeClassifier(ccp_alpha=0.1)

↓

Much simpler tree

↓

Potential underfitting